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Name Solution

April 26, 2008 Saturday

Department _____

SID _____

University Physics
Midterm Examination
Kuang Yaming Honors School, Nanjing University

Physical Constants

Gravitational constant

$$G = 6.67 \times 10^{-11} \text{ m}^3 / \text{s}^2 \cdot \text{kg}$$

Standard gravity acceleration

$$g_n = 9.806 \text{ m/s}^2$$

Select five out of following six problems.

(1) (20pts) A block of mass m moves on a horizontal surface. The friction force acted on the block is $f = \alpha + \beta v^2$ where α and β are constants. The initial velocity of the block is v_0 .

(a) Write the equation of motion of the block.

(b) After what time interval does the block come to rest?

(c) What distance has then been covered by the block?

Solution: (a) The equation of motion reads

$$m \frac{dv}{dt} = -f = -\alpha - \beta v^2.$$

(b) The above equation can be rewritten as

$$\frac{dv}{\frac{\alpha}{\beta} + v^2} = -\frac{\beta}{m} dt.$$

Integrating the both sides of above equation, we have

$$\int_{v_0}^0 \frac{dv}{\frac{\alpha}{\beta} + v^2} = -\frac{\beta}{m} \int_0^t dt \quad \text{or} \quad \sqrt{\frac{\beta}{\alpha}} \left(\arctan 0 - \arctan \sqrt{\frac{\beta}{\alpha}} v_0 \right) = -\frac{\beta}{m} t.$$

Thus, we have

$$t = \frac{m}{\sqrt{\alpha\beta}} \arctan \sqrt{\frac{\beta}{\alpha}} v_0.$$

(c) Rewrite the equation of motion as

$$m \frac{dv}{dt} = m \frac{dv}{dx} \frac{dx}{dt} = mv \frac{dv}{dx} = -\alpha - \beta v^2 \quad \text{or} \quad \frac{mv dv}{\alpha + \beta v^2} = -dx.$$

Integrating both sides of above equation

$$\frac{m}{2\beta} \int_{v_0}^0 \frac{2\beta v dv}{\alpha + \beta v^2} = - \int_0^x dx,$$

we finally obtain

$$x = -\frac{m}{2\beta} \int_{v_0}^0 \frac{2\beta v dv}{\alpha + \beta v^2} = -\frac{m}{2\beta} \int_{v_0}^0 \frac{d(\alpha + \beta v^2)}{\alpha + \beta v^2} = -\frac{m}{2\beta} \ln \frac{\alpha}{\alpha + \beta v_0^2} = \frac{m}{2\beta} \ln \frac{\alpha + \beta v_0^2}{\alpha}.$$

(2) (20pts) A particle of mass m moves in one dimension has a potential energy

$$U(x) = U_0 \left[2 \left(\frac{x}{a} \right)^2 - \left(\frac{x}{a} \right)^4 \right]$$

where U_0 and a are positive constants.

(a) Find the force $F(x)$, which acts on the particle.

(b) Find the positions of stable and unstable equilibrium.

(c) What is the minimum speed the particle must have at the origin to escape to infinity?

Solution: (a) $F(x) = -\frac{dU(x)}{dx} = -\frac{4U_0 x}{a^2} \left(1 - \frac{x^2}{a^2} \right).$

(b) From $F(x) = 0$, we obtain $x = 0, x = \pm a$.

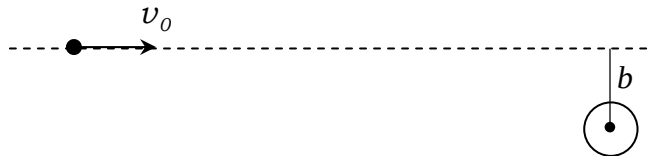
For $x = 0$, $U''(0) > 0$. It is a stable equilibrium position.

For $x = \pm a$, $U''(\pm a) < 0$. They are unstable equilibrium positions.

(c) $U_{\max} = U(\pm a) = U_0$, $U(0) = 0$. Then, we have

$$\frac{1}{2} m v_{\text{esc}}^2 + 0 = U_0 \quad \text{or} \quad v_{\text{esc}} = \sqrt{\frac{2U_0}{m}}.$$

(3) (20pts) A space probe heads towards a very distant planet of radius R and very large mass M with initial velocity v_0 and impact parameter b .



(a) Find the maximum value of the impact

parameter b for which the probe will hit the planet.

(b) What is the velocity of the probe when it just grazes the planet?

Solution: (a) Assume that the velocity of the probe at the instant that the probe barely grazes the planet is v_f .

Conservation of angular momentum: $mv_f R = mv_0 b_{\max}.$

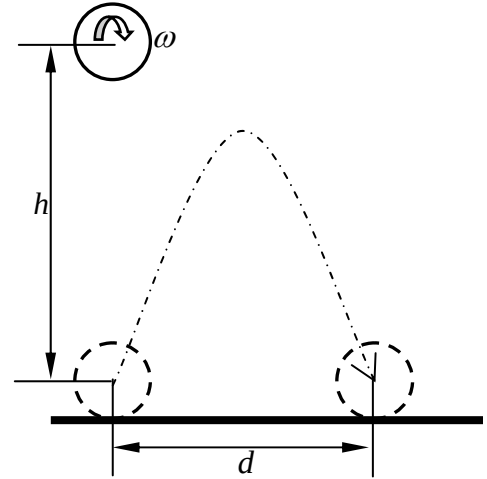
Conservation of mechanical energy: $\frac{1}{2} m v_0^2 = \frac{1}{2} m v_f^2 - \frac{GMm}{R}.$

Eliminating v_f from above equations, we have

$$\frac{1}{2} v_0^2 \left(\frac{b_{\max}^2}{R^2} - 1 \right) = \frac{GM}{R} \quad \text{or} \quad b_{\max} = R \sqrt{1 + \frac{2GM}{v_0^2 R}}.$$

(b) $v_f = \frac{v_0 b_{\max}}{R} = v_0 \sqrt{1 + \frac{2GM}{v_0^2 R}}.$

(4) (20pts) If a ball of mass m is dropped from rest at a height h from a horizontal ground, it bounces back to a height of kh where $0 < k < 1$ after it hits the ground. If the ball is dropped again from the same height with a sufficiently large initial angular velocity ω , it bounces and drops to the ground a distance d apart from the point it first hits the ground, as shown in the accompanying figure. Assume h and d are much larger than the radius of the ball.



(a) What is the impulse I on the ball from the collision with the ground when it is dropped from rest?

(b) What is the kinetic frictional coefficient μ between the surface of the ball and the ground?

Solution: (a) The velocity of the ball before collision is

$$v_1 = -\sqrt{2gh}.$$

The velocity of the ball after collision is $v_2 = -\sqrt{2gkh}$. Thus, the impulse acted on the ball is

$$I = p_f - p_i = m(v_2 - v_1) = m\sqrt{2gh}(1 + \sqrt{k}).$$

(b) Using the friction law $f = \mu N$, we have

$$\Delta p_x = \int f dt = \int \mu N dt = \mu \int N dt = \mu I.$$

The horizontal velocity of the ball gained due to the friction is

$$v_x = \frac{\Delta p_x}{m} = \frac{\mu I}{m} = \mu \sqrt{2gh}(1 + \sqrt{k}).$$

Using $d = v_x t$, $t = \frac{2v_2}{g}$, we have

$$d = \frac{2v_x v_2}{g} = \frac{2\sqrt{2gkh}\mu\sqrt{2gh}(1 + \sqrt{k})}{g} = 4\mu h\sqrt{k}(1 + \sqrt{k}),$$

or

$$\mu = \frac{d}{4h(\sqrt{k} + k)}.$$

(5) (20pts) A uniform solid cylinder of mass m and radius R is at rest on a frictionless horizontal surface. The cylinder is struck horizontally at a point on its circumference that is a height h above the surface. Immediately after this instantaneous hitting, the cylinder begins to roll without slipping at a velocity v_0 . And the cylinder then encounters an inclined plane that makes an angle α with the horizontal. The surface of the inclined plane is also frictionless.



(a) Find the rotational inertia of cylinder about its center of mass. Find the height h of the hitting point.

(b) Does the cylinder continue to roll without slipping upon the inclined plane? Why?

(c) What is the maximum height $H_{\max}$ that the cylinder can reach on the inclined plane?

Solution: (a) The rotational inertia of the cylinder about its center of mass is

$$I_C = \frac{1}{2}mR^2. \text{ (Please refer to the textbook or the lecture notes for the detailed calculation.)}$$

Condition for rolling without slipping is $v_0 = \omega_0 R$. Thus, if the impulse acted on the cylinder is J , we have

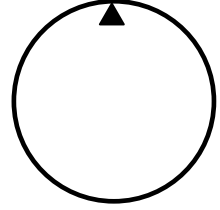
$$I_c \omega_0 = J(h - R) = mv_0(h - R) \quad \text{or} \quad h = \frac{I_c \omega_0}{mv_0} + R = \frac{\frac{1}{2}mR^2 \omega_0}{\omega_0 R} + R = \frac{3}{2}R.$$

(b) Because there is no friction acted, the angular velocity of the cylinder will be unchanged while the translational velocity of the center of the mass will be reduced. Therefore, the cylinder will begin to slip upon the inclined plane.

(c) Since the rotational kinetic energy K_r is unchanged, we have

$$K_t = \frac{1}{2}mv_0^2 = mgH_{\max} \quad \text{or} \quad H_{\max} = \frac{v_0^2}{2g}.$$

(6) (20pts) As shown in the figure, a thin circular hoop is suspended at its top in a way so that it can oscillate freely in the vertical plane of the hoop. The mass of the hoop is M and the radius of it is R .



(a) Write down the equation of rotational motion of the hoop.

(b) Find the period of small oscillations about its equilibrium position.

(c) If the hoop oscillates in the direction perpendicular to its plane, what is the period of small oscillations?

Solution: (a) The rotational inertia about the suspension point is

$$I = I_c + mR^2 = mR^2 + mR^2 = 2mR^2.$$

Thus, the equation rotational motion is

$$I\ddot{\theta} = -mgR \sin \theta \quad \text{or} \quad \ddot{\theta} + \frac{g}{2R} \sin \theta = 0.$$

(b) For small oscillations, we have $\sin \theta \approx \theta$. The equation of motion can be rewritten as

$$\ddot{\theta} + \frac{g}{2R} \theta = 0.$$

Thus, $\omega^2 = \frac{g}{2R}$ and $T = 2\pi \sqrt{\frac{2R}{g}}.$

(c) The rotational inertia about the suspension point in this case is

$$I' = I_c + mR^2 = \frac{1}{2}mR^2 + mR^2 = \frac{3}{2}mR^2.$$

And the equation of rotational motion for small oscillations is

$$\ddot{\theta} + \frac{2g}{3R} \theta = 0.$$

Thus, $\omega'^2 = \frac{2g}{3R}$ and $T' = 2\pi \sqrt{\frac{3R}{2g}}.$